

Operating System Concepts - Mini Textbook

1. Introduction to Operating Systems

An Operating System (OS) is system software that manages computer hardware, software resources, and provides services for computer programs. Examples include Windows, Linux, macOS, and Android.

2. Process Management & Scheduling

A process is a program in execution. The OS handles process creation, execution, and termination. Scheduling algorithms (FCFS, SJF, Round Robin, Priority Scheduling) decide the execution order of processes to optimize CPU utilization and responsiveness.

3. Threads & Concurrency

Threads are lightweight processes that share resources within a process. Concurrency control ensures multiple processes/threads run without conflicts. Multithreading improves system performance by running tasks in parallel.

4. Memory Management & Virtual Memory

The OS allocates and manages memory for processes. Techniques include paging, segmentation, and swapping. Virtual memory allows execution of processes larger than physical memory using demand paging.

5. File Systems

The OS provides a file system to organize, store, and access data on storage devices. It manages directories, file access methods, and implements permissions for security.

6. Input/Output Systems

I/O management controls communication between hardware devices and applications. It uses device drivers, buffering, spooling, and caching to ensure efficient data transfer.

7. Deadlocks

A deadlock occurs when processes are waiting indefinitely for resources held by each other. Deadlock handling strategies include prevention, avoidance (Banker's algorithm), detection, and recovery.

8. Security & Protection

OS security ensures that data and resources are protected from unauthorized access. Techniques include authentication, access control, encryption, and auditing.

9. Distributed Systems

A distributed OS manages a collection of independent computers and makes them appear as a single coherent system. It provides transparency, fault tolerance, and resource sharing.

10. Recent Trends in OS

Modern OS developments include virtualization, cloud-based OS, real-time OS, and mobile operating systems focused on security, scalability, and efficiency.